Detecting AI-Generated Images: A Comparative Study of CNN and Vision Transformer Models with Grad-CAM Interpretability

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DS 5500 Data Capstone

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GitHub Repo Link

March 24, 2026

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1 Purpose of the Methodology

The goal of this project is not only to classify AI-generated versus human-created images, but also to understand how models make these decisions. To achieve this, we adopt a comparative and interpretability-driven methodology combining convolutional neural networks (CNNs), vision transformers (ViTs), and post-hoc explanation techniques.

First, we train two representative deep learning models: a ResNet (CNN-based architecture) and a Vision Transformer (ViT) on a balanced dataset of AI-generated and authentic images. These models represent two fundamentally different paradigms in computer vision: CNNs focus on local spatial features such as edges and textures, while transformers model global relationships through attention mechanisms. Comparing these architectures allows us to investigate whether different inductive biases lead to different detection behaviors.

Model comparison is necessary because prior research shows that detection performance can vary significantly depending on architecture and training setup. For example, Moebner and Adel demonstrate that detection systems may rely on fragile high-frequency artifacts, leading to performance drops under image transformations . Similarly, Guillaro et al. highlight that generalization issues often stem from models learning spurious dataset-specific cues rather than robust forensic signals . By comparing ResNet and ViT, we aim to identify whether one model is more robust, generalizable, or interpretable than the other.

Beyond performance metrics such as accuracy and loss, we incorporate Grad-CAM (Gradient-weighted Class Activation Mapping) to analyze model interpretability. This allows us to visualize which regions of an image contribute most to a model's prediction. Prior work suggests that humans and AI rely on different cues—humans focus on semantic content, while models often rely on textures or noise patterns . By applying Grad-CAM, we aim to validate whether our models exhibit similar behavior and to identify specific visual artifacts associated with AI-generated images.

2 Problem Statement

2.1 Defining the Problem

This project addresses a supervised machine learning problem in the domain of computer vision, specifically a binary image classification task. The objective is to classify images as either AI-generated or human-created based on their visual content. Given an input image, the model must learn discriminative features that enable it to distinguish synthetic images from authentic ones.

The dataset used in this project consists of approximately 79,950 images with an

equal distribution between AI-generated and human-created classes, providing a balanced setting for supervised learning . Each authentic image is paired with a corresponding AI-generated counterpart, which helps control for differences in content and ensures that the classification task focuses on subtle visual distinctions rather than high-level semantic differences.

In addition to classification, the project incorporates interpretability analysis as a secondary objective. This extends the problem beyond prediction to include understanding and explaining model behavior, making it both a classification and model analysis task within machine learning.

2.2 Significance

The problem of detecting AI-generated images has become increasingly important due to the rapid advancement of generative models, which are capable of producing highly realistic visual content. As these models improve, the visual differences between synthetic and authentic images become more difficult to detect, posing significant challenges for individuals and institutions that rely on the authenticity of digital media.

As outlined in the project proposal, the proliferation of AI-generated images raises concerns about misinformation, manipulation, and the erosion of trust in visual content . For example, synthetic images can be used to create misleading narratives or fabricated evidence, which can influence public opinion and decision-making processes. This makes reliable detection methods essential for applications such as social media moderation, journalism, and forensic analysis.

From a research perspective, this problem remains challenging because existing detection systems often lack robustness and interpretability. Velasquez et al. show that humans frequently misclassify AI-generated images, with a substantial portion of synthetic images perceived as real . At the same time, many machine learning models achieve high accuracy but provide limited insight into how their decisions are made. Other work has shown that detection models may rely on superficial artifacts rather than meaningful semantic inconsistencies, which can reduce their effectiveness when faced with new or unseen generative techniques .

This project is therefore significant because it addresses both performance and interpretability. By comparing different model architectures and analyzing their decision-making processes, the study contributes to a better understanding of how AI-generated images can be detected and what visual cues are most informative. These insights are critical for developing more robust, transparent, and trustworthy detection systems that can adapt to the rapidly evolving landscape of generative AI.

3 Data Collection and Preparation

3.1 Data Sources

The dataset used in this project is the AI vs Human-Generated Images Dataset from Kaggle. It contains both authentic images and AI-generated images produced by modern generative models. The dataset was chosen because it is large, balanced, and designed specifically for comparing real and synthetic visual content. No additional data collection or synthetic data generation was performed.

3.2 Data Description

The dataset consists of approximately 79,950 images, evenly split between AI-generated and human-created classes . Each image is labeled as either real (label 0) or synthetic (label 1), forming a binary classification task.

The images cover a variety of categories, including objects, scenes, and human subjects, which increases the diversity and difficulty of the task. Additionally, the dataset includes paired real and AI-generated images with similar content, allowing models to focus on subtle visual differences rather than obvious semantic cues.

Although the dataset is balanced, it may still contain biases related to the specific generative models used, which could affect how well the trained models generalize to other types of AI-generated images.

For development and reproducibility on local and HPC hardware, a 5,000-image working subset is used throughout this project. This subset preserves the original 50/50 class balance of the full benchmark.

Table 1: Summary of the dataset.

Attribute	**Value**
Total Samples	79,950
Actual Samples Used	5,000
Number of Classes	2
Training / Validation / Test Split	60% / 20% / 20%
Class Distribution	AI-Generated: 50%, Authentic: 50%

3.3 Preprocessing Steps

Several preprocessing steps were applied to prepare the data for training deep learning models. First, the dataset was cleaned by verifying image paths and removing corrupted or unreadable files to ensure data quality. The data was then split into training, validation, and test sets using controlled random sampling to maintain class balance across all subsets.

All images were resized to a fixed resolution compatible with both ResNet and Vision Transformer architectures. During preprocessing, images were resized to 256×256 pixels and center-cropped to 224×224 for evaluation. Pixel values were converted from the integer range [0, 255] to floating-point values in [0, 1], followed by normalization using ImageNet statistics (mean = [0.485, 0.456, 0.406], standard deviation = [0.229, 0.224, 0.225]) to match the pretrained backbone requirements.

To improve model robustness and reduce overfitting, data augmentation was applied during training. The augmentation pipeline included random resized cropping to 224×224 pixels, random horizontal flipping with probability 0.5, and color jittering ($\pm 20\%$ brightness, contrast, saturation, and $\pm 10\%$ hue). These transformations increase data diversity and encourage the model to learn invariant and generalizable features rather than memorizing specific patterns.

No random transformations were applied during validation and testing to ensure deterministic and reproducible evaluation.

No explicit feature selection was performed, as both ResNet and Vision Transformer architectures automatically learn hierarchical feature representations from raw image inputs. Additionally, no specialized outlier removal beyond initial data cleaning was conducted, as deep learning models are generally robust to minor data variations when trained on sufficiently large datasets.

Overall, these preprocessing steps ensure that the dataset is clean, standardized, and suitable for robust training and reliable evaluation of AI-generated image detection models.

4 Selection of Machine Learning or LLM Models

4.1 Model Consideration

Traditional machine learning methods—including Decision Trees, Support Vector Machines (SVMs), Random Forests, and gradient-boosted models such as XGBoost—were considered during the project scoping phase. While these models are computationally efficient and well-understood, they are not well-suited for raw image classification. They rely on hand-crafted feature extraction (e.g., HOG descriptors, frequency-domain statistics, or texture features), which introduces significant manual design effort and often fails to generalize across the diverse range of AI-generated image artifacts. Moreover, their representational capacity is limited compared to deep neural networks. Prior work in image forensics consistently demonstrates that transfer learning with deep models significantly outperforms shallow classifiers, even on moderately sized datasets. For these reasons, traditional ML approaches were not pursued further.

Among deep learning models, this project focuses on two representative architectures

from distinct families: convolutional neural networks (CNNs) and Vision Transformers.

ResNet-50[1] is selected as the CNN representative. CNNs have been the dominant paradigm in computer vision for over a decade and are highly effective in transfer learning settings. ResNet-50, in particular, provides a well-established and thoroughly benchmarked baseline, with pretrained ImageNet features known to transfer robustly to downstream classification tasks.

ViT-B/16[2] is selected as the Transformer-based representative. Unlike CNNs, Vision Transformers process images as sequences of fixed-size patches using self-attention mechanisms. This architecture introduces a distinct inductive bias: global attention rather than local convolution. As a result, ViT models are capable of capturing long-range spatial dependencies and global inconsistencies that may be indicative of AI-generated image artifacts.

The final candidate models were selected based on three criteria: (1) strong pretrained representations for transfer learning, (2) architectural diversity to capture complementary feature types (CNN vs Transformer), and (3) computational feasibility given limited training time and resources.

Comparing these architectures allows us to assess whether global attention provides measurable advantages over convolutional feature extraction in this task.

Table 2: Model Architectures and Pretraining

Architecture	**Family**	**Pretrained on**
ResNet-50	CNN (residual)	ImageNet-1K
ViT-B/16	Vision Transformer	ImageNet-1K

Table 3: Parameter Count and Linear Probe Configuration

Architecture	**Backbone Parameters (Frozen)**	**Linear Probe (Trained)**
ResNet-50	∼25.6 M	4,098 (=2,048 × 2 + 2)
ViT-B/16	∼86.6 M	1,538 (=768 × 2 + 2)

4.2 Final Model Selection

Rather than selecting a single model, both architectures are retained as parallel experimental tracks. The goal is explicitly comparative: to determine which feature representation—convolutional or attention-based—is more effective for AI-generated image detection under identical training conditions.

For both models, a *linear probing* setup is adopted: the pretrained backbone is frozen, and only a lightweight classification head is trained. This ensures that performance

differences are attributable to the quality of the learned feature representations rather than differences in fine-tuning dynamics.

For **ResNet-50**, the 2,048-dimensional feature vector obtained from global average pooling of the final convolutional layer serves as input to the classifier. For **ViT-B/16**, the 768-dimensional CLS token representation—aggregated through self-attention across all image patches—is used as the input feature.

This controlled experimental design enables a fair comparison between CNN-based local feature extraction and Transformer-based global attention in detecting artifacts present in AI-generated images.

4.3 Evaluation Metrics

The models are evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score.

Given the perfectly balanced class distribution, the following metrics are computed after every validation epoch and once on the held-out test set:

Table 4: Evaluation Metrics and Interpretations

Metric	Interpretation
Accuracy	Fraction of all predictions that are correct; unbiased because classes are balanced.
ROC AUC	Area under the Receiver Operating Characteristic curve; threshold-independent measure of discrimination ability; 1.0 = perfect, 0.5 = random.
Precision	Of all images predicted as AI-generated, what fraction truly are? (Controls false alarms).
Recall	Of all AI-generated images, what fraction are correctly detected? (Controls missed detections).
F1-Score	Harmonic mean of precision and recall; a single balanced summary of both error types.
Confusion Matrix	Full breakdown of true positives, true negatives, false positives, and false negatives.

ROC AUC is used as the primary checkpoint-selection criterion (alongside validation loss) because it is threshold-agnostic and robust to the model's calibration quality. Accuracy is reported as the primary summary metric for human-readable comparison.

5 Model Development and Training

5.1 Model Architecture and Configuration

Both models are trained using transfer learning, where ImageNet-1K pretrained backbones are reused as feature extractors and adapted to the binary classification task through a linear probing setup. In this configuration, the pretrained backbone is frozen and only a lightweight classification head is trained. This design allows performance differences to be attributed primarily to the quality of the learned feature representations rather than to differences in full-model fine-tuning dynamics.

5.1.1 ResNet-50

The ResNet-50 backbone is initialized with the standard ImageNet-1K V1 pretrained weights released by `torchvision`. The architecture consists of an initial 7×7 convolutional stem, followed by four residual stages containing 3, 4, 6, and 3 bottleneck blocks, respectively, a global average pooling layer, and an original 1,000-class fully connected head.

For this project, the original classification head is replaced with a binary linear classifier mapping a 2,048-dimensional feature vector to 2 output logits. Under the linear probing protocol, all backbone parameters are frozen, totalling approximately 25.5 million parameters, while only the new head is updated during training. The trainable head contains 4,098 parameters ($2{,}048 \times 2 + 2$), representing approximately 0.016% of the full model.

Table 5 summarizes the main configuration used for the ResNet-50 linear probe.

Table 5: ResNet-50 linear probe configuration

Component	**Configuration**
Pretrained weights	ImageNet-1K V1
Input resolution	224×224
Feature dimension	2,048
Classification head	Linear layer ($2{,}048 \rightarrow 2$)
Frozen backbone parameters	$\sim$25.5M
Trainable head parameters	4,098
Batch size	64
Head learning rate	3×10^{-4}
Weight decay	1×10^{-2}
Label smoothing	0.1
Maximum epochs	20
Early stopping patience	5
Gradient clipping norm	1.0
Mixed precision	Enabled
Random seed	42

5.1.2 Vision Transformer (ViT-B/16)

The ViT-B/16 backbone is initialized with ImageNet-1K V1 pretrained weights. The model divides a 224×224 image into a grid of $14 \times 14 = 196$ non-overlapping patches, each of size 16×16 pixels. Each patch is embedded into a 768-dimensional vector. A learnable CLS token is prepended to the patch sequence, resulting in 197 input tokens.

The sequence is processed by 12 transformer encoder layers. Each layer contains a multi-head self-attention block with 12 attention heads (head dimension 64) and a feed-forward network with hidden dimension 3,072, connected through residual paths and layer normalization. The final CLS token representation is used as the image-level feature vector.

For binary classification, the original multiclass head is replaced with a linear classifier mapping 768 dimensions to 2 output logits. Under the linear probing setup, approximately 86.6 million backbone parameters are frozen and only the new head is trained. The trainable head contains 1,538 parameters.

Table 6 summarizes the configuration used for the ViT-B/16 linear probe.

Table 6: ViT-B/16 linear probe configuration

Component	**Configuration**
Pretrained weights	ImageNet-1K V1
Input resolution	224×224
Patch size	16×16
Number of patches	196
Token sequence length	197 (including CLS token)
Number of encoder layers	12
Number of attention heads	12
Embedding dimension	768
MLP hidden dimension	3,072
Classification head	Linear layer ($768 \to 2$)
Frozen backbone parameters	$\sim$86.6M
Trainable head parameters	1,538
Batch size	64
Head learning rate	3×10^{-4}
Weight decay	1×10^{-2}
Label smoothing	0.1
Maximum epochs	20
Early stopping patience	5
Gradient clipping norm	1.0
Mixed precision	Enabled
Learning-rate floor ($\eta_{\min}$)	1×10^{-5}
Warmup epochs	0
Random seed	42

5.1.3 Planned Two-Stage Fine-Tuning Strategy

Although the current experiments use linear probing only, a progressive fine-tuning strategy is planned for the next iteration. The motivation is that a head-only probe may underutilize the representational capacity of large pretrained backbones, especially for ViT-B/16.

The planned strategy consists of two stages:

1. **Stage 1: Linear probing.** The pretrained backbone remains frozen and only the classification head is trained.

2. **Stage 2: Progressive unfreezing.** For ViT-B/16, the last two transformer encoder blocks will be unfrozen; for ResNet-50, the final residual stage (`layer4`) will be unfrozen. A differential learning rate will be used, where the backbone receives a learning rate $30\times$ smaller than the head (1×10^{-5} vs. 3×10^{-4}), in order to adapt the pretrained representation without excessively disrupting previously learned features.

5.2 Training Process

The dataset contains 5,000 images and is split into three non-overlapping subsets using stratified sampling to preserve the 50/50 class balance between human-created and AI-generated images:

- **Training set:** 3,000 images (60%), used for gradient updates.
- **Validation set:** 1,000 images (20%), used at the end of each epoch for model selection and early stopping.
- **Test set:** 1,000 images (20%), used only once after training is complete to report final held-out performance.

No hyperparameter choices or checkpoint decisions are made using test-set results, ensuring that the final test metrics remain unbiased estimates of generalization performance.

Loss function. Both models are trained using cross-entropy loss with label smoothing ($\varepsilon = 0.1$). Label smoothing replaces hard one-hot targets with soft targets, reducing overconfidence and serving as an additional regularizer.

Optimizer. Training uses the AdamW optimizer. The classification head receives the main learning rate of 3×10^{-4}, while any unfrozen backbone parameters in later fine-tuning experiments receive a reduced learning rate of 1×10^{-5} to protect pretrained representations from large updates.

Learning-rate schedule. A cosine annealing schedule is used to reduce the learning rate smoothly over training, with a minimum floor of 1×10^{-5}. For the linear probing phase, warmup is omitted because only the small classification head is trained and optimization is expected to remain stable.

Regularization and overfitting control. Several techniques are used to reduce overfitting and improve training stability:

- **Weight decay:** 1×10^{-2}
- **Label smoothing:** 0.1
- **Early stopping:** patience of 5 epochs based on validation loss
- **Gradient clipping:** global norm clipped to 1.0

- **Frozen pretrained backbone:** reduces the number of trainable parameters and limits overfitting on the smaller target dataset

If validation loss does not improve for 5 consecutive epochs, training is terminated and the best checkpoint is restored.

Mixed precision training. Automatic mixed precision (AMP) is enabled on GPU to improve computational efficiency while maintaining numerical stability. Forward passes are computed in FP16, while gradient updates are stabilized through FP32 scaling and accumulation.

Training configuration. For both models, the maximum training duration is 20 epochs, the batch size is 64, and the random seed is fixed at 42 for reproducibility.

Hardware and software environment. Experiments are implemented in PyTorch using `torchvision` pretrained models and are executed on NVIDIA GPU hardware with CUDA support. ResNet-50 experiments are conducted on an NVIDIA T4 GPU, while ViT-B/16 experiments are run on an NVIDIA V100 GPU due to its higher memory capacity and computational throughput requirements. Mixed-precision training is enabled where supported to improve efficiency.

5.3 Hyperparameter Tuning

In the current iteration, hyperparameters are selected using principled defaults drawn from prior transfer-learning and Vision Transformer fine-tuning literature, rather than from exhaustive automated search. This choice provides a strong and reproducible baseline while keeping the initial comparison between ResNet-50 and ViT-B/16 controlled and computationally feasible.

Table 7 summarizes the main hyperparameters used in the current experiments and the rationale for each choice.

Table 7: Hyperparameter settings and rationale

Hyperparameter	**Value**	**Rationale**
Head learning rate	3×10^{-4}	Standard AdamW starting point for linear probing and commonly recommended in transfer-learning practice.
Backbone learning rate	1×10^{-5}	Set 30× lower than the head learning rate to reduce catastrophic forgetting.
Weight decay	1×10^{-2}	Provides moderate L2-style regularization and is consistent with standard AdamW defaults.
Batch size	64	Balances GPU memory usage with stable gradient estimates for 224×224 images.
Label smoothing	0.1	Reduces overconfidence and improves generalization for small classification heads.
Gradient clipping norm	1.0	Stabilizes optimization and limits rare large gradient updates.
Maximum epochs	20	Sufficient for rapid convergence under a linear probing setup.
Early stopping patience	5	Prevents unnecessary training once validation performance plateaus.

A more systematic hyperparameter search is planned for the next stage of the project. Candidate search methods include grid search for discrete variables such as batch size and number of unfrozen blocks, and Bayesian optimization for continuous variables such as learning rate and weight decay. The primary search space will include head learning rate, backbone learning rate, batch size, weight decay, and the number of unfrozen layers or blocks.

6 Evaluation and Comparison

6.1 Performance Metrics

The performance of the models in this project is evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and ROC AUC. Accuracy measures overall correctness, while precision and recall assess how well the model distinguishes between AI-generated and human-created images. The F1-score provides a balanced measure of these two, and ROC AUC evaluates the model's ability to separate the two classes across different decision thresholds.

For the ResNet-50 model, evaluation on the test set shows strong performance across all metrics. The model achieved a test accuracy of 90.20% and a ROC AUC of 0.9662, indicating excellent discriminative ability. Precision, recall, and F1-score are all 0.9020,

demonstrating balanced performance between detecting AI-generated and authentic images. The confusion matrix further supports this, with relatively low misclassification rates (50 false positives and 48 false negatives), suggesting that the model performs consistently across both classes.

During training, the model also demonstrated stable convergence. The final validation accuracy reached 92.30% with a ROC AUC of 0.9749, indicating strong generalization from training to validation data. The relatively small gap between validation and test performance suggests that overfitting was effectively controlled through preprocessing and data augmentation techniques.

The Vision Transformer (ViT-B/16) model achieved a lower test accuracy of 85.90% and a ROC AUC of 0.9294. Compared to ResNet-50, this indicates weaker performance in distinguishing between AI-generated and human-created images under the current training setup. One possible explanation is that ViT models typically require larger datasets and longer training to fully leverage their global attention mechanisms, whereas CNNs like ResNet are more effective at capturing local artifacts that are highly relevant for this task.

Table 8: Test Performance comparison of evaluated models.

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**ROC AUC**
ResNet-50	0.9020	0.9020	0.9020	0.9020	0.9662
ViT-B/16	0.8590	0.8590	0.8590	0.8590	0.9294

Overall, the comparison shows that ResNet-50 outperforms ViT-B/16 in both accuracy and discriminative ability. This suggests that, for AI-generated image detection, local texture-based features play a more critical role than global contextual relationships. As a result, ResNet-50 was selected as the stronger model in this study. However, the difference in architectural behavior also highlights an important insight: while CNNs excel at detecting artifact-level cues, transformer-based models may offer complementary advantages if trained at a larger scale or combined with other approaches.

6.2 Interpretability Comparison

In addition to quantitative metrics, qualitative analysis using Grad-CAM provides further insight into how each model arrives at its predictions. Examining the same real image processed by both models reveals distinct attention patterns. ResNet-50 produces activations that are broadly distributed across background regions, particularly focusing on natural textures such as the wheat field's surface and peripheral environmental details. This suggests the model relies on localized, low-level features — edges, textures, and fine-grained pixel-level inconsistencies — when distinguishing real from AI-generated images. With a low AI probability of 0.0495, ResNet-50 confidently identifies the image as real,

likely leveraging the organic textural variation present in the scene. This aligns with prior research indicating that CNN-based architectures excel at detecting high-frequency artifacts and subtle local inconsistencies commonly found in generated images, though this reliance on surface-level patterns could limit robustness against novel generative methods. In contrast, ViT's Grad-CAM activations concentrate directly on the human subject — particularly the torso, limbs, and head region — indicating that the model attends to global, semantically meaningful structures rather than local textures. Despite correctly classifying the image as real, ViT assigns a notably higher AI probability of 0.4010, reflecting lower confidence compared to ResNet-50. This pattern suggests that ViT evaluates the structural coherence and anatomical plausibility of the subject as a whole, consistent with the self-attention mechanism's ability to capture long-range spatial dependencies across image patches. However, this global reasoning approach may explain ViT's lower overall accuracy, as it appears less sensitive to the fine-grained textural cues that often serve as reliable indicators of AI generation.

Figure 1 presents the Grad-CAM visualizations for both models on the same input image, illustrating these contrasting attention strategies.

(a) ResNet-50 Grad-CAM (AI prob: 0.0495).

(b) ViT Grad-CAM (AI prob: 0.4010).

Figure 1: Grad-CAM comparison between ResNet-50 (top row) and ViT-B/16 (bottom row) on the same real image. ResNet-50 attends to diffuse background textures, while ViT focuses on the central human subject, reflecting their respective local vs. global feature extraction strategies.

References

[1] He, K., Zhang, X., Ren, S., Sun, J. (2016). Deep Residual Learning for Image Recognition. *CVPR*, 770-778.

[2] Dosovitskiy, A., et al. (2021). An Image is Worth 16x16 Words. *ICLR*.